Community Evacuation Resource Matcher

Ramblin' Pathfinders - Georgia Institute of Technology & Accenture Song

Path: Accelerating Equitable Evacuations

Solution Statement: Our tool connects vulnerable communities needing evacuation help with nearby volunteers.

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The Problem

During wildfire evacuations, elderly, disabled and car-free residents face the greatest risk, not just from lack of resources but also from lack of coordination

Identifying the Gap

Key insights from CalFire, SLO OES, Santa Clara Firesafe Council:

1) Helpful to know **who** needs help, **what** help they need, and **where** they are

2) Community evacuation assistance is **highly effective**

Disconnected Neighborhoods

Residents in high-need tracts have limited ways to signal what help they need or find neighbors who can assist

Unmatched Resources

Hard for helpers to identify which communities to prioritize and what resources they need

No Real-Time Routing

Without location-aware matching, aid is distributed unevenly

Data Landscape

- Data Sources:
  - Watch Duty
  - CalFire
  - American Community Survey (Census Data)
  - Census Tract Data

- Key Variables
  - % of households with elderly (1+ persons 65 years and over)
  - % of population with disability (civilian noninstitutionalized population)
  - % of households without car
  - Census tract GEO IDs (Census, Watch Duty, CalFire)

- Data Cleaning and Pre-Processing
  - Combined census tract FIPS codes, tract geometry, and demographic data into one dataset, removed tracts with missing demographic info
  - Chose **Butte, Shasta, and Riverside** counties due to having 50+ census tracts and high demographic variability

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Technical Approach Overview

Step 1

Designed algorithm to match helpers to highest-need census tracts using census demographic data

Step 2

Built full-stack matching engine with LLM-powered request parsing and live map dashboard

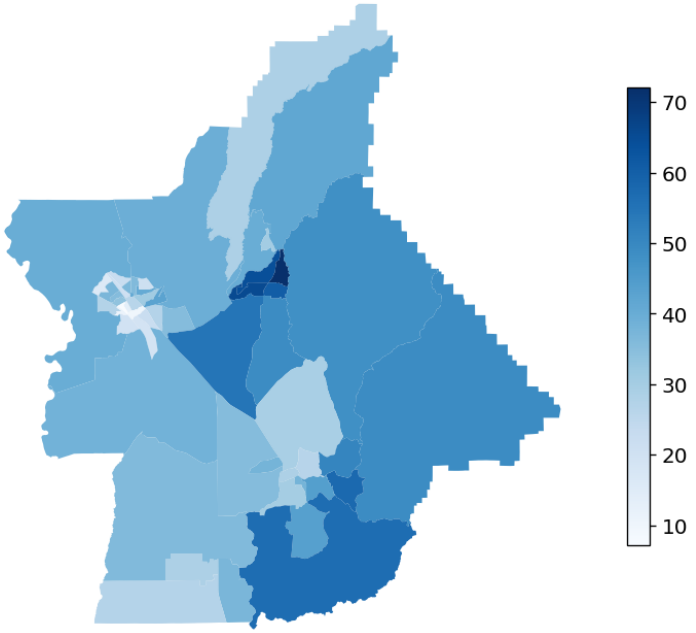
Step 3

Used GenAI to iteratively refine matching logic, UI, and capabilities into production-ready prototype

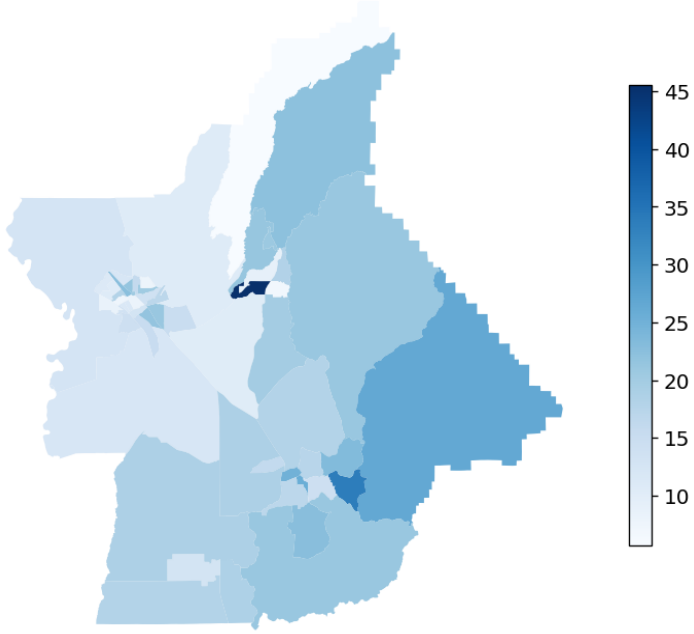
Visualizing County Demographics

Percentage distributions in Butte County: Elderly, Disability, No-Vehicle

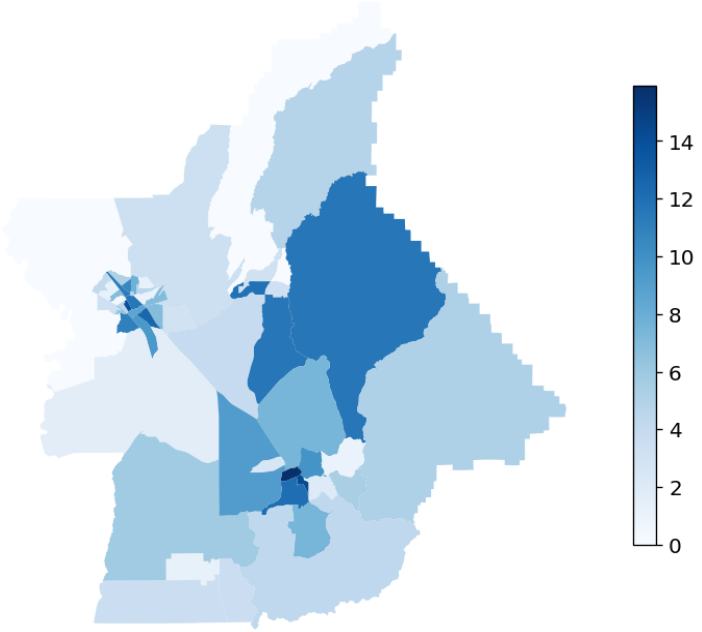
Percentage of Elderly People



Percentage of People with Disabilities



Percentage of People without Vehicles



Visuals help us check for demographic heterogeneity to allow for informative resource and service prioritization.

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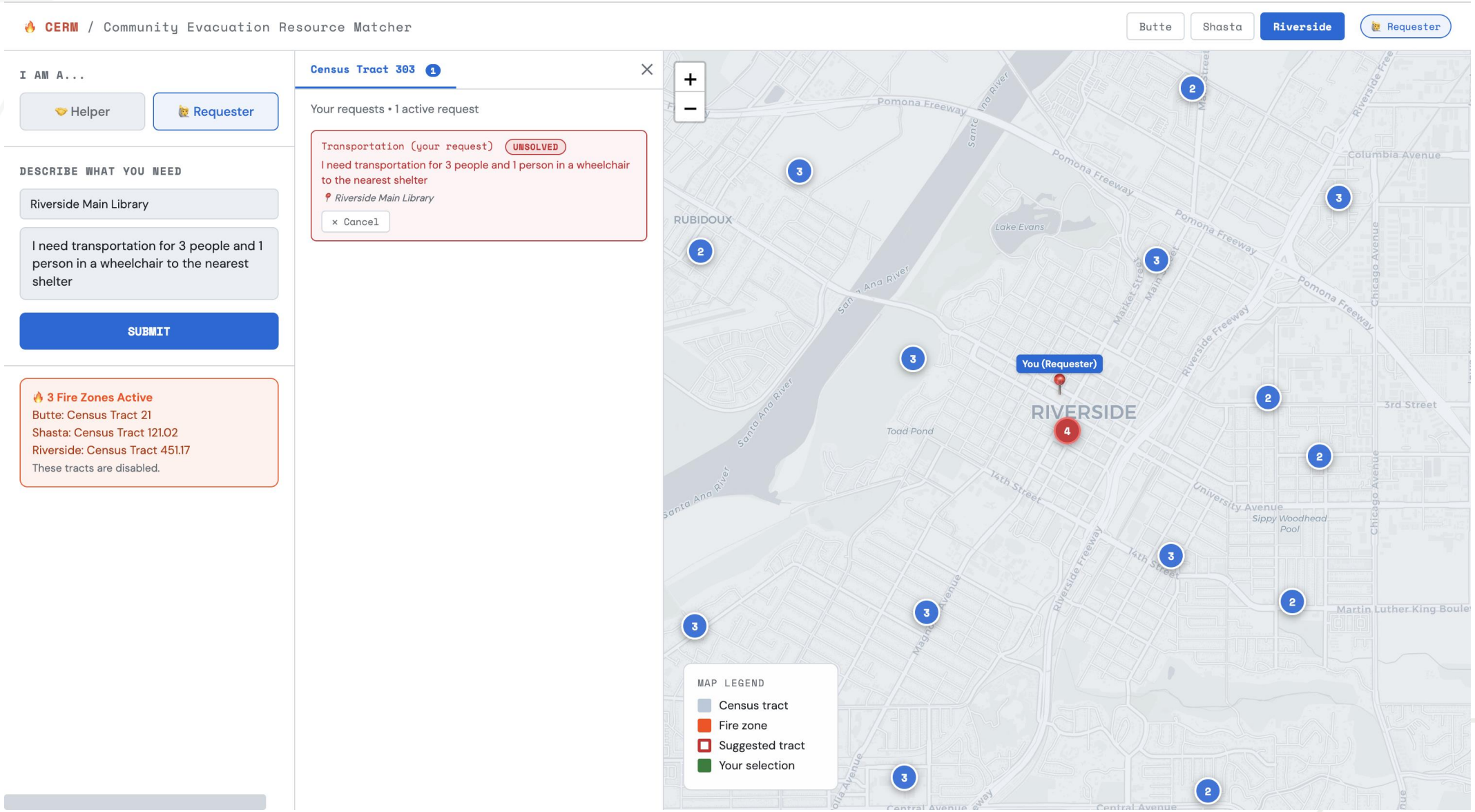
Community Evacuation Resource Matcher (CERM)



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Building CERM

Matching Engine Algorithm

1. Helpers and Requesters describe what they can offer or need
2. DeepSeek LLM parses inputs in real time into structured tags
3. Rule-based matching algorithm matches helpers to census tracts based on **proximity, vulnerability fit, request category match, and request volume**

Component	Weight	Description
Proximity	35%	Distance between the helper and the tract centroid, with a decay half-life of ~10 km
Vulnerability Fit	25%	Alignment between the helper's target beneficiaries and the tract's vulnerability flags
Request Category Match	25%	Proportion of unresolved requests that match the helper's offered services
Request Volume	15%	Normalized count of active requests within the tract

Tech Stack

1. Frontend
   - Single-page app in HTML, CSS, and JavaScript
   - Leaflet.js for interactive mapping
   - Hosted on GitHub Pages
2. Backend
   - Python data pipeline for census data cleaning and spatial processing.
   - LLM calls proxied through a Vercel serverless function

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AI and Agentic Integration

Integrating GenAI Into Solution

- **Core of Matching Engine**
  - DeepSeek LLM parses helper and requester inputs into structured need and resource tags
- **Human-Centered UX**
  - Generates plain-language explanations of why a helper is matched to specific tracts and what requests they can fulfill

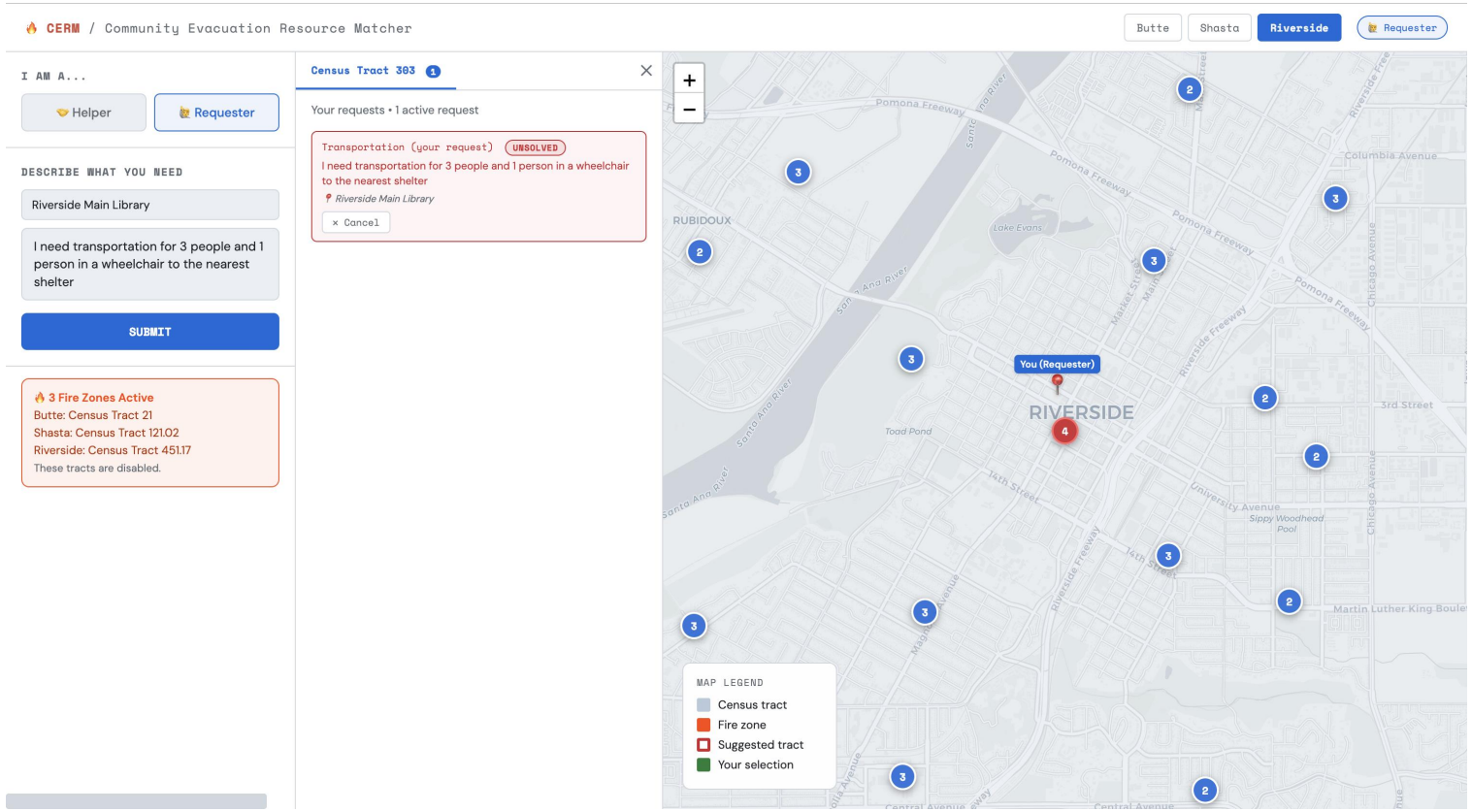
Using GenAI During Development

- **Rapid Prototyping**
  - Claude served as a core development partner for data pipeline design, debugging, and UI iteration
- **Accelerated Iteration**
  - Enabled rapid pivoting after expert interviews with fire professionals, translating feedback into product changes in hours

Results and Validation

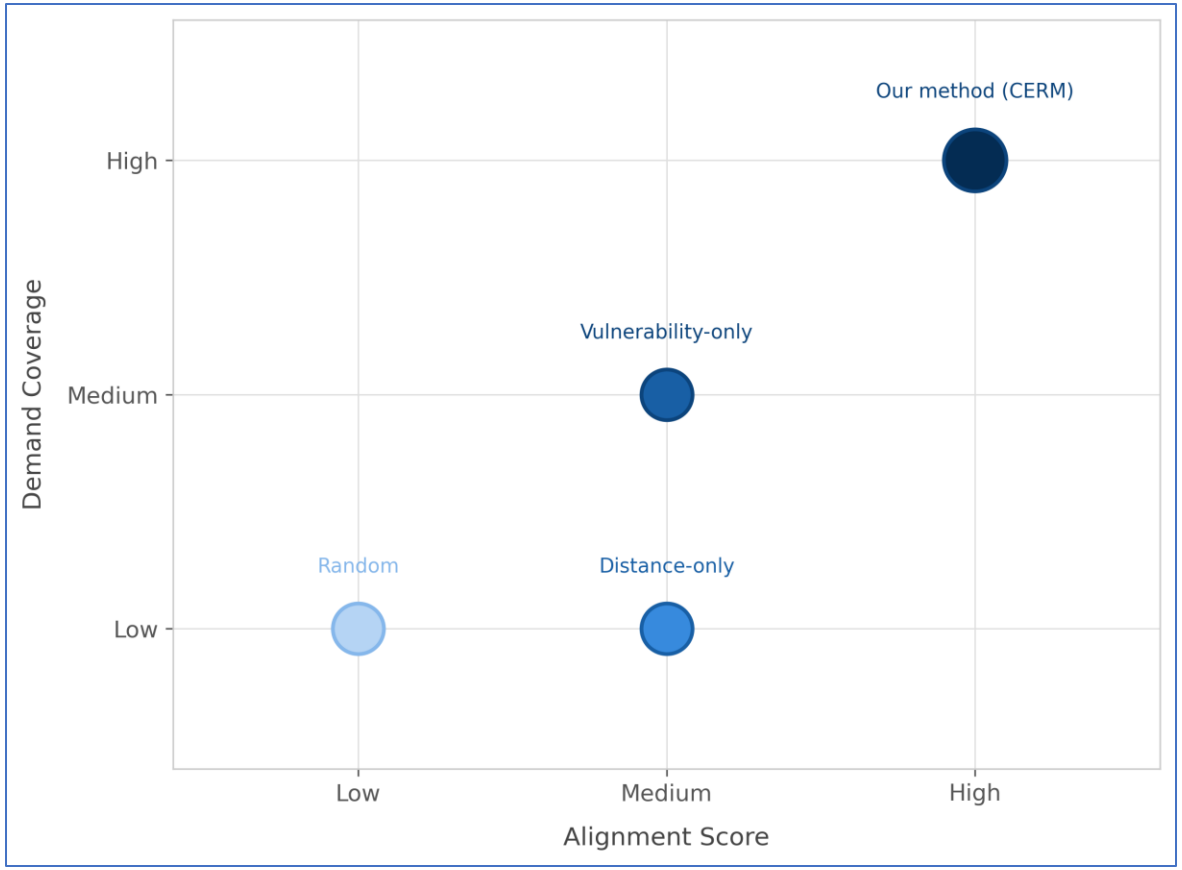
Helper workflow: https://youtu.be/wNPNUmJdeUQ

Requester workflow: https://youtu.be/nXE0r4IYQjE



Results and Validation

Why CERM over Other Services?



Our method takes distance and vulnerability of demographics of areas into account, beating the other single-factor methods or random matching baseline.

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Results and Validation

Category	Metric	Definition
Performance Metric	Evacuation Efficiency	$\frac{\text{Avg. Evacuation Time Difference}}{\text{Avg. Evacuation Time Before Deployment}} \times 100$
	Request Alignment Score	$\frac{\text{Helpers assigned to tracts with category} - \text{matched request}}{\text{Total helpers assigned to tracts with request}} \times 100$
	Demand Coverage	$\frac{\text{Requests resolved within T hours}}{\text{Total distinct requests submitted within T hours}} \times 100$
	Vulnerable Group Demand Coverage	Demand Coverage calculated within each of the vulnerable demographic groups: 1) Elderly, 2) People with Disabilities, 3) People without Vehicles
	Mean Accepted Recommendation Rank	Average Recommendation Rank of the Tracts that got Accepted. Ideally this should be closer to 1, and if higher it means our recommendation algorithm is not helpful for the users
Adoption Metric	User Signups	Number of users signed up (QoQ Growth)
	Active Users During Fire	Number of users that utilized our tool during a fire

Evolution and Pivots

Idea 1: Fire Safety Organizations

Target Audience: Fire Safety Professionals

Concept: Resource allocation + preparedness visualization

Why We Pivoted: Existing tools cover operational needs, resource allocation before a fire isn't as viable

Idea 2: Resource Locater for Civilians

Target Audience: Civilians

Concept: Resource-locator for community needs

Why We Pivoted: Identified bigger gap in community help / mutual aid

Final Idea: Community Evacuation Resource Matching Tool

1. Fills gap in community coordination
2. Uses GenAI & matching algorithms → improves evacuation efficiency

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Limitations and Next Steps

Limitations

- Simulated demand
- Static weights for distance, demographic, matching
- No traffic constraints
- LLM parsing errors

Assumptions

- Requesters will be responsible updating their request in the tool
- People/ orgs are willing to help without putting themselves in harm's way
- Users will make reasonable requests and provide reasonable aid
- Vulnerable populations (elderly, disabled, no vehicles) are more likely to use this tool as requesters

Next Steps

- Add direct chat feature between the helper and requester the helper chose
- Include real-time data (traffic, fire spread) to update suggestions for helpers
- Identity authentication
- Authorize tool to access users' precise location
- Multiple language compatibility
- Voice-to-text abilities

Thank You

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